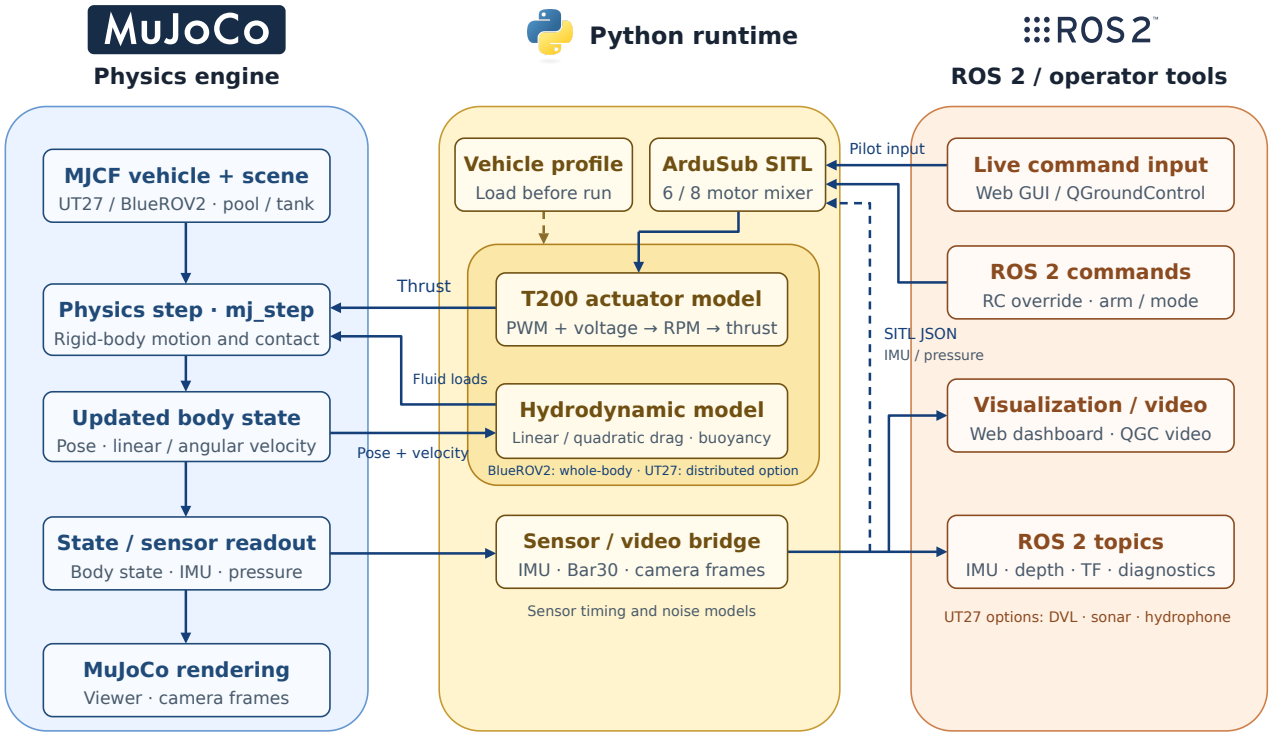

MuJoCo
Physics engine
Python runtime
ROS 2
ROS 2 / operator tools
MJCF vehicle + scene
UT27 / BlueROV2 · pool / tank
Physics step · mj_step
Rigid-body motion and contact
Updated body state
Pose · linear / angular velocity
State / sensor readout
Body state · IMU · pressure
MuJoCo rendering
Viewer · camera frames
Vehicle profile
Load before run
ArduSub SITL
6 / 8 motor mixer
Thrust
T200 actuator model
PWM + voltage → RPM → thrust
Fluid loads
Hydrodynamic model
Linear / quadratic drag · buoyancy
BlueROV2: whole-body · UT27: distributed option
Pose + velocity
Sensor / video bridge
IMU · Bar30 · camera frames
Sensor timing and noise models
Pilot input
Live command input
Web GUI / QGroundControl
ROS 2 commands
RC override · arm / mode
SITL JSON
IMU / pressure
Visualization / video
Web dashboard · QGC video
ROS 2 topics
IMU · depth · TF · diagnostics
UT27 options: DVL · sonar · hydrophone